

QINGDAO, China

WMO: 548570

Lat: 36.0628N

Lon: 120.3348E

Elev: 253

StdP: 14.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	19.5	-2.1	5.0	29.5	1.6	6.0	30.4	25.6	27.3	23.2	28.1	6.1	340	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	91.5	75.7	89.2	74.9	86.3	74.3	80.9	86.0	79.7	84.3	78.6	82.8	9.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	153.2	83.7	78.6	149.9	83.0	77.2	143.0	81.7	45.1	86.0	43.7	84.5	42.5	83.6	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	20.1	17.8	DB	10.8	96.6	3.4	3.3	8.4	99.0	6.4	100.9	4.6	102.8	2.1	105.2
			WB	9.3	83.3	3.4	1.3	6.9	84.2	4.9	84.9	3.0	85.6	0.5	86.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	31.0	35.5	45.0	56.1	66.5	73.6	79.9	79.8	72.4	61.3	47.2	35.1	(6)
(7)		DBStd	18.07	6.09	6.67	6.94	5.88	5.19	4.06	3.83	3.88	4.77	6.24	7.58	6.48	(7)
(8)		HDD50	1801	589	405	186	14	0	0	0	0	0	3	141	463	(8)
(9)		HDD65	4425	1054	825	620	271	43	1	0	0	5	146	533	927	(9)
(10)		CDD50	4387	0	1	31	199	513	708	926	924	671	354	58	2	(10)
(11)		CDD65	1532	0	0	0	5	91	259	461	459	226	31	0	0	(11)
(12)		CDH74	11486	0	0	1	56	579	1398	3851	4209	1289	102	1	0	(12)
(13)		CDH80	3423	0	0	0	11	145	419	1254	1312	272	10	0	0	(13)
(14)	Wind	WSAvg	8.3	7.9	8.4	9.6	10.1	9.3	8.8	7.9	7.3	6.9	7.4	8.0	8.0	(14)
(15)	Precipitation	PrecAvg	28.3	0.4	0.6	0.8	1.3	2.2	3.1	6.8	6.0	3.3	1.7	1.1	0.5	(15)
(16)		PrecMax	54.0	1.8	2.0	3.7	4.1	6.3	9.9	19.6	19.0	15.8	5.9	5.2	1.5	(16)
(17)		PrecMin	12.0	0.0	0.0	0.0	0.0	0.1	0.1	0.8	1.1	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	9.9	0.5	0.5	0.7	0.9	1.4	2.2	3.9	4.2	3.1	1.5	1.3	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	56.8	70.2	80.4	89.6	93.1	95.4	93.6	89.3	80.8	69.7	56.9	(19)
(20)		MCWB	41.8	45.5	53.2	61.5	65.8	69.9	77.6	78.1	70.6	64.8	58.6	50.4	(20)	
(21)		2%	DB	46.6	51.9	64.4	73.8	84.0	88.0	91.7	91.0	85.8	76.8	65.4	53.2	(21)
(22)		MCWB	40.7	42.4	49.8	57.5	63.9	68.8	77.5	77.2	70.1	62.2	55.9	47.3	(22)	
(23)		5%	DB	44.2	48.5	59.2	69.9	80.3	84.5	89.3	88.1	82.8	73.7	62.8	49.8	(23)
(24)		MCWB	38.5	40.5	47.5	55.5	62.2	68.5	76.6	76.6	68.7	61.7	55.3	44.2	(24)	
(25)		10%	DB	41.4	46.0	55.5	66.2	76.8	81.3	86.2	86.2	80.7	71.5	60.5	46.7	(25)
(26)		MCWB	36.3	39.6	44.9	53.3	60.9	67.9	76.4	76.0	68.5	61.3	53.7	41.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	48.7	56.2	64.5	70.1	77.4	83.0	82.7	78.2	69.7	62.2	52.9	(27)
(28)		MDCB	47.4	53.7	64.8	74.1	80.4	84.2	90.1	87.6	82.0	74.0	65.4	54.9	(28)	
(29)		2%	WB	42.0	44.8	52.2	61.0	67.4	74.6	81.0	81.1	75.7	67.3	59.6	48.2	(29)
(30)		MDCB	45.1	48.7	60.8	69.7	77.6	80.4	86.5	85.8	80.0	72.1	63.2	51.7	(30)	
(31)		5%	WB	39.4	42.5	49.5	58.4	65.6	72.7	79.7	80.0	73.6	65.4	57.0	44.8	(31)
(32)		MDCB	42.8	46.4	56.5	66.0	74.6	78.6	84.6	84.6	78.5	70.7	61.5	48.2	(32)	
(33)		10%	WB	36.9	40.4	46.9	56.0	63.8	71.1	78.6	79.1	71.8	63.5	54.0	41.9	(33)
(34)		MDCB	40.9	44.2	53.0	62.7	72.0	76.7	83.0	83.2	77.4	69.2	59.6	46.0	(34)	
(35)	Mean Daily Temperature Range		MDBR	16.3	17.1	18.2	18.9	18.9	16.5	13.5	13.2	16.3	17.8	17.1	16.3	(35)
(36)		5% DB	MCDBR	18.8	19.8	22.6	23.6	23.8	21.4	17.1	14.9	18.3	19.1	18.8	19.2	(36)
(37)			MCWBR	13.0	12.4	12.3	11.8	10.5	8.5	6.9	6.5	8.1	9.8	11.8	13.9	(37)
(38)			MCDDBR	16.4	16.7	20.4	20.5	20.6	16.3	13.7	12.2	13.9	15.6	15.3	16.4	(38)
(39)		5% WB	MCWBR	12.9	12.1	12.4	12.0	10.8	8.0	6.8	6.2	7.9	9.7	11.7	13.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.503	0.557	0.603	0.617	0.638	0.737	0.721	0.654	0.575	0.599	0.567	0.506	(40)	
(41)		taud	1.797	1.713	1.652	1.670	1.672	1.539	1.625	1.744	1.853	1.711	1.730	1.804	(41)	
(42)		Ebn at Noon	202	208	214	221	220	198	200	210	219	194	180	190	(42)	
(43)		Edh at Noon	53	65	75	77	78	89	81	71	60	64	56	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	774	990	1411	1636	1767	1646	1491	1478	1348	1106	808	695	(44)	
(45)		RadStd	77	115	110	97	133	123	155	168	115	110	90	59	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	-2.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	N/A

Nomenclature: See separate page